INCH-POUND
MIL-PRF-12883/46C
28 June 1996
SUPERSEDING
MIL-S-12883/46B
20 December 1989

PERFORMANCE SPECIFICATION SHEET

SOCKET, PLUG-IN ELECTRONIC COMPONENTS, FOR RELAYS, 3-POLE, 10 AMPERES (MIL-R-6106 AND MIL-R-83536)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.

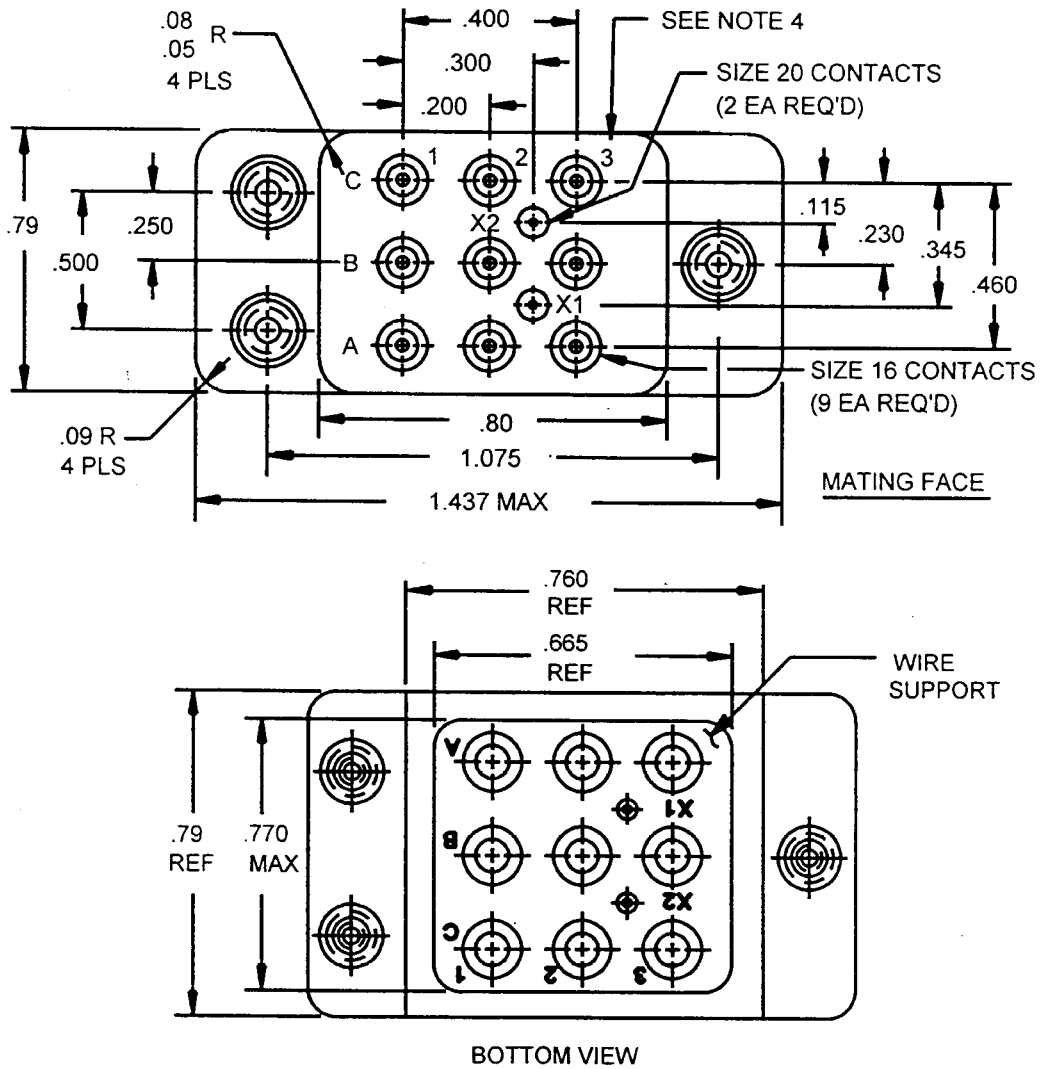
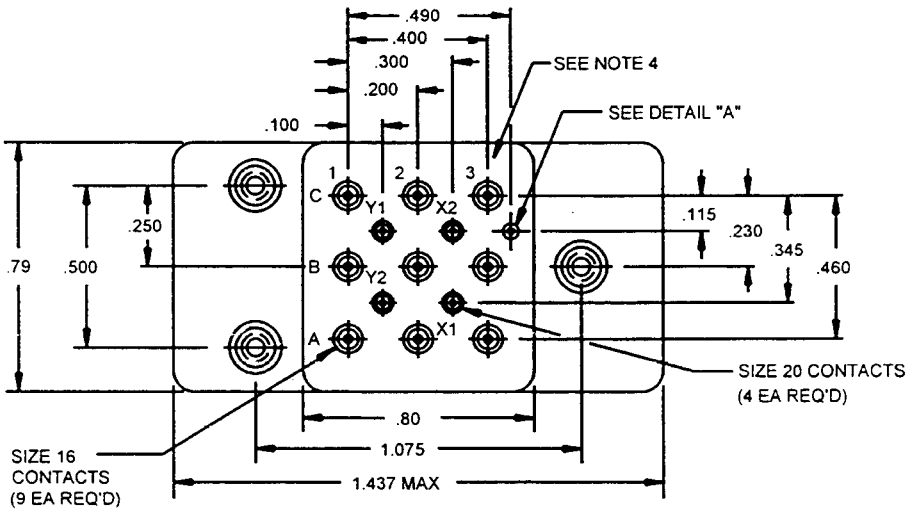


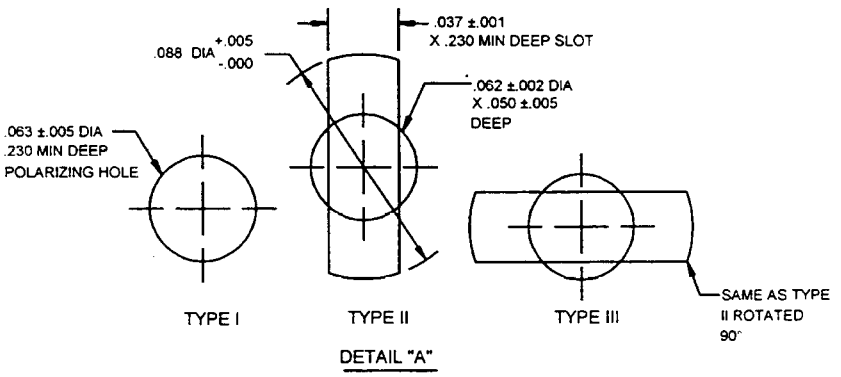
FIGURE 1. Socket configurations (-01).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

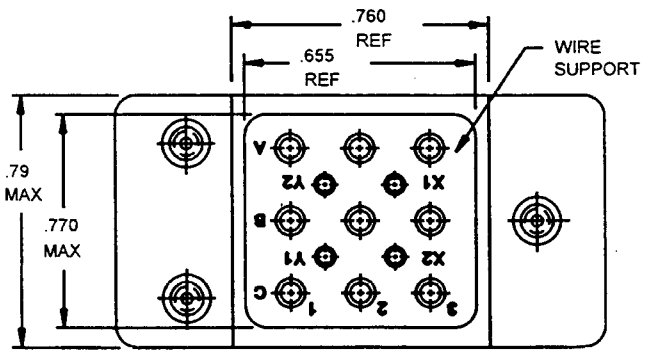
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MATING FACE



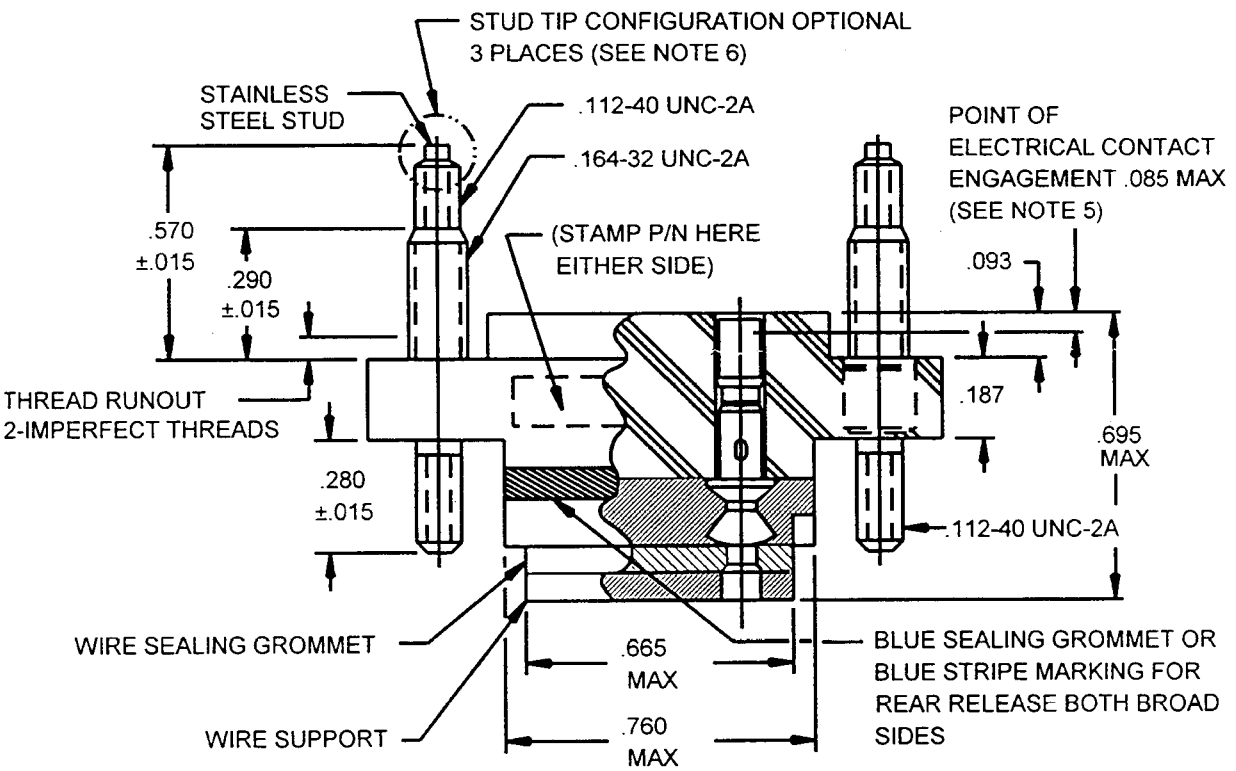
DETAIL "A"



BOTTOM VIEW

FIGURE 2. Socket configurations (-02, 03, and -04).

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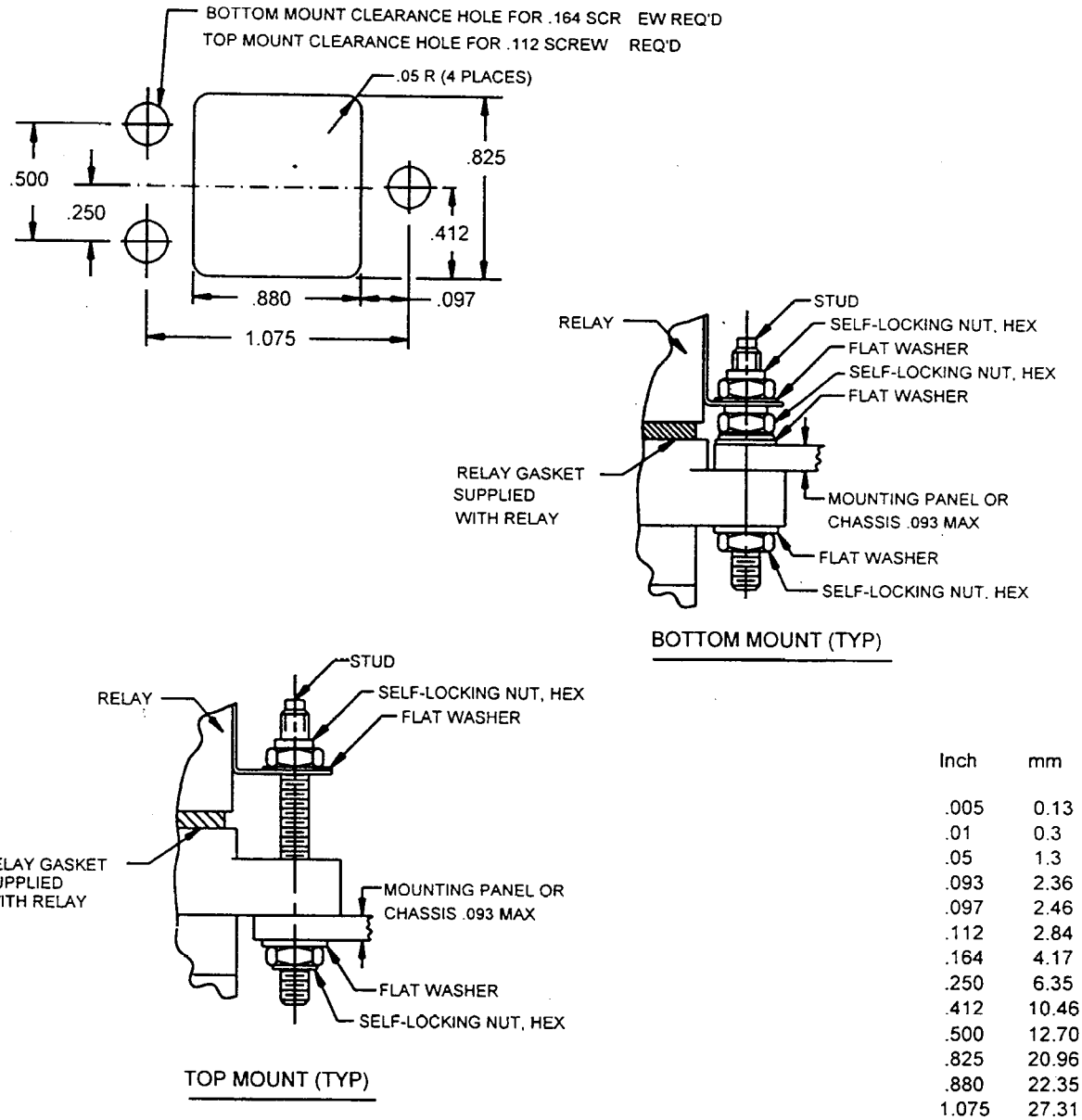
Inch	mm	Inch	mm	Inch	mm
.001	0.03	.09	2.3	.345	8.76
.002	0.05	.093	2.36	.400	10.16
.005	0.13	.100	2.54	.460	11.68
.01	0.25	.112	2.84	.490	12.45
.015	0.38	.115	2.92	.500	12.70
.037	0.94	.164	4.17	.570	14.48
.05	1.3	.187	4.75	.665	16.89
.050	1.27	.200	5.08	.695	17.65
.062	1.57	.230	5.84	.760	19.30
.063	1.60	.250	6.35	.770	19.56
.08	2.0	.280	7.11	.79	20.1
.085	2.16	.290	7.37	.80	20.3
.088	2.24	.300	7.62	1.075	27.31
				1.437	36.50

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 inch (0.25 mm) for two place decimals.
4. Marking shall be characters which are molded .035 inch (0.89 mm) minimum, ink marking optional; see MIL-STD-1285.
5. Point of electrical contact engagement from mating face of socket insulator to the socket contact.
6. The configuration of the stud tip shall be optional provided it meets all the requirements of the specification.

FIGURE 3 Socket configurations (–01, 02, -03, AND -04).

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Inch	mm
.005	0.13
.01	0.3
.05	1.3
.093	2.36
.097	2.46
.112	2.84
.164	4.17
.250	6.35
.412	10.46
.500	12.70
.825	20.96
.880	22.35
1.075	27.31

)TES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerances are ±.005 inch (0.13 mm) for three place decimals and ±.01 (0.25 mm) for two place decimals.

FIGURE 4. Socket mounting (hardware and panel).

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REQUIREMENTS:

QUALIFICATION:

Test gage: A mating relay in accordance with table I shall be used as a test gauge or mating connector.

Reference material: (see 3.1.1 of MIL-PRF-12883)

Insulator and wire support: Diallyl phthalate, in accordance with MIL-M-14, type SDG-F or any glass filled thermoplastic material in accordance with MIL-P-46179, or an approved equivalent material manufactured to a commonly accepted industry standard.

Color: Material color shall be optional providing that the color provides a contrasting background for the blue sealing grommet/blue color bands indicating rear release contacts.

Contact: In accordance with MIL-C-39029/101 (see table I), or approved equivalent contacts manufactured to a commonly accepted industry standard.

Grommet: Silicon rubber in accordance with ZZ-R-765.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Design, construction and physical dimensions: Design, construction and physical dimensions shall be as specified in figures 1, 2, 3, 4, and table I.

TABLE I. Dash number and characteristics.

Dash number	Contact size mating end wire barrel		Number of contacts	Contact designation	Mating relay 1/	Polarization type
01	16	16	9	M39029/101-554	MIL-R-83536/21-019, /21-022, /22-022	None
	20	20	2	M39029/101-553		
02	16	16	9	M39029/101-554	M6106/40-002	I
	20	20	4	M39029/101-553		
03	16	16	9	M39029/101-554		II
	20	20	4	M39029/101-553		
04	16	16	9	M39029/101-554		III
	20	20	4	M39029/101-553		

1/ Reference MIL-R-83536 for supersession data on MIL-R-6106 relays.

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Performance:

Electrical:

Insulation resistance: 1000 megohms minimum.

Dielectric withstanding voltage:

Test voltage at sea level shall be 1,250 V rms.

Test voltage at 80,000 feet (24,384 Meters) shall be 500 V rms.

Environmental:

Temperature range: Operating temperature range shall be -70°C to +125°C.

Wire sealing: A resilient grommet is permanently bonded to the wire entry face of the socket so as to provide moisture sealing capabilities, over the following range of finished wire;

Contact	Wire diameter range
M39029/101-553	.040 min - .083 max (1.02 min - 2.11 max mm)
M39029/101-554	.053 min - .103 max (1.35 min - .103 max mm)

Mechanical:

Contacts: Contact shall be in accordance with table I. Contacts shall be crimp removable type, rear release and accept relay pins size 16 (.0610 through .0630 inch (1.55 through 1.60 mm)) and size 20 (.039 through .041 inch (.099 through 1.04 mm)).

Insertion and withdrawal force: The insertion and withdrawal forces shall be as specified in table II.

TABLE II. Insertion and withdrawal force.

M12883/46	-01	-02 through -04
Insertion force (max)	19.1 lbf (85 newton)	21.4 lbf (95.2 newton)
Withdrawal force (min)	1.2 lbf (5.3 newton)	1.3 lbf (5.78 newton)
After 10 insertions and withdrawals, before vibration Insertion force (max)	23 lbf (102.3 newton)	25.8 lbf (114.8 newton)
Withdrawal force (min)	.92 lbf (4.1 newton)	1.0 lbf (4.45 newton)
After vibration Insertion force (max)	23 lbf (102.3 newton)	25.8 lbf (114.8 newton)
Withdrawal force (min)	.92 lbf (4.1 newton)	.99 lbf (4.40 newton)

Durability: Shall be in accordance with MIL-PRF-12883.

Vibration (sinusoidal): In accordance with MIL-STD-202, method 204, test condition G, or an equivalent industry standard.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 Hz and 3,000 Hz.

b. Except that the procedure of method 201 of MIL-STD-202 may be applied during 10 Hz to 55 Hz band of the vibration frequency range.

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Vibration (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, or an equivalent industry standard, with a test duration 15 minutes. The mating relay shall be used as the test gage.

Shock (mechanical): In accordance with MIL-STD-202, method 213, condition C, or an equivalent industry standard, except peak value shall be 200 g's.

Torque: Relay sockets and hardware shall be subjected to testing torque as specified in table III. Sockets shall be installed in mounting panel when test torque is applied. No physical damage will be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged.

TABLE III. Torque requirements (installed in panel conditions).

Thread size	Torque			
	Testing		Installation	
	Inch-pounds	Newton-meters	Inch-pounds	Newton-meters
.112-40	8 + 1 - 0	0.90 + .11 - 0	4 ± 1	0.45 ± .11
.164-32	20 + 1 - 0	2.20 + .11 - 0	15 ± 1	1.70 ± .11

Socket, contact tools: Tools for crimping the electrical contacts, inserting and removing the contacts from the socket body is specified in table IV. Approved equivalent industry standard tools may be used where appropriate.

TABLE IV. Contact tools.

Nomenclature	Part or Identifying Number (PIN)
Crimp tool	M22520/7-01
Positioners	M22520/7-12 and -13
Insertion and removal tool	M81969/16-01 and -02

Weight: The socket, hardware, and electrical contacts shall have a maximum weight of .055 pounds (25 grams).

Mounting hardware: The mounting hardware shall allow mounting the socket above or below the panel orchassis (see figure 4), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal .197 inch (5.00 mm) spacing between socket surface and relay mounting flange, regardless of mounting configuration.

Mounting hardware shall be supplied with the relay socket and shall consist of the following:

Self locking nuts .112-40 (6 each) (.224 max dia x .176 max height).

Flat washers .112 (6 each) (.220 max O.D. x .021 max thickness).

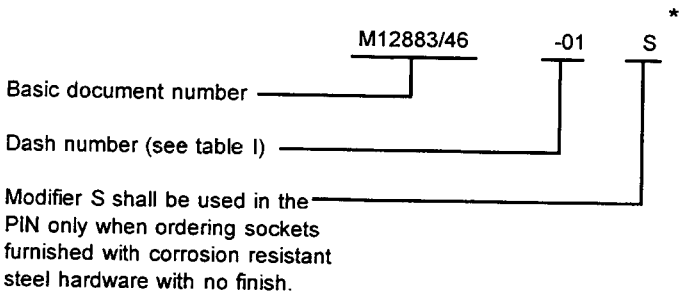
Self locking nuts .164-32 (3 each) (.290 max dia x .190 max height).

Flat washers .164 (3 each) (.290 max O.D. x.019 max thickness).

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PIN: The PIN shall be marked on the socket body as shown in the example (see figure 3). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



* For future acquisition of these sockets as of the effective date of revision B, 20 December 1989, parts identified with an "S: modifier shall be corrosion resisting steel (CRS), and parts without an "S" modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Packaging: Hardware, socket insulator, and contacts shall be packaged in separate packages within a common container.

Revision letters are not used to denote changes due to the extensiveness of the changes.

The Government PIN, specified in table VI, supersedes the following commercial part numbers.

TABLE VI. Supersession and cross reference.

Active Government PIN	Superseded manufacturers part number	
	CAGE 58982	CAGE 99699
M12883/46-01	RSE116747	SE310-1010
M12883/46-02	RSE116749	TBD
M12883/46-03	RSE116751	TBD
M12883/46-04	RSE116753	TBD

CONCLUDING MATERIAL

Custodian:
Army - CR
Navy - EC
Air Force - 85

Review activity:
Army - AR, AS, AT, AV, MC, ME
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - ES

(Project 5935-3990-02)

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